

Neonatal Intensive Care Workflow Analysis Informing NEC-Zero Clinical Decision Support Design

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Decision support in neonatal ICUs is needed, especially for prevention and risk awareness of the devastating complication of necrotizing enterocolitis, a major cause of emergency surgery among fragile infants. The purpose of this study was to describe the current clinical workflow and sociotechnical processes among clinicians for necrotizing enterocolitis risk awareness, timely recognition of symptoms, and treatment to inform decision support design. A qualitative descriptive study was conducted. Focus groups were held in two neonatal ICUs (five groups in Unit A and six in Unit B). Transcripts were analyzed using content analysis and compared with field notes. Clinicians (N = 27) included nurses (37%), physicians (30%), neonatal nurse practitioners (19%), and other staff (16%). Workflow processes differed for nurses (who see necrotizing enterocolitis signs and notify providers to order diagnostic tests and treatments) and providers (who receive notification of necrotizing enterocolitis concern and then decide how to act). Clinicians desired (1) a necrotizing enterocolitis-relevant dashboard to support nutrition tracking and necrotizing enterocolitis recognition; (2) features to support decision-making (eg, necrotizing enterocolitis risk and adherence scoring); (3) breast milk tracking and feeding clinical decision support; (4) tools for necrotizing enterocolitis surveillance and quality reporting; and (5) general EHR optimizations to improve user experience.

KEY WORDS: Clinical decision support, Necrotizing enterocolitis, NEC-Zero, Neonatal intensive care, Workflow mapping

In neonatal intensive care, clinicians keep watch for the very serious condition of necrotizing enterocolitis (NEC). Necrotizing enterocolitis is an intestinal complication that generally develops 2-5 weeks after birth,¹ especially in fragile infants who are born with very low birth weight (<1500 g) or are fragile due to birth defects such as congenital heart disease.^{2,3} In mild cases, NEC can be treated with medical care consisting of intravenous antibiotics, gastric decompression, and bowel rest. When NEC worsens, it can be treated with a bedside drain and often requires exploratory laparotomy to remove dead bowel tissue.⁴ If very severe, the intestine can perforate, and infection will spread systemically throughout the infant's body, which has rare and underwhelming ability to overcome the inflammatory insults from widespread sepsis. Low situational awareness and risk recognition of an infant developing NEC could lead to delayed diagnosis and slow treatment.⁵⁻⁷ Because of the nature of neonatal ICU (NICU) care, some infants who develop NEC will need to be transferred to another hospital to have surgery and the bowel can perforate and the infant may die while being transported. For these reasons, strategies are needed to promote effective surveillance and prevention activities, foster risk awareness, and support timely recognition and technology that could help improve the consistency of those strategies.⁵⁻⁷

BACKGROUND

Although prevention evidence to reduce NEC occurrence is available and of low cost,⁸ integrating prevention recommendations into clinical practice is not consistent, and tracking adherence is not simple. The long-term goal of this program of research is to implement clinical decision support (CDS) tools to support consistent NEC prevention, NEC risk awareness, and timely recognition.⁹ To be successful and useful to clinicians, tools need to fit the clinician's workflow in the NICU to avoid cognitive dissonance, minimize unintended consequences, and, ultimately, offer an advantage over the current ways of managing high-risk infants. In many settings, with the broad implementation of EHR systems, clinicians can be subjected to workflows that deliver misaligned, suboptimal experiences. When this is the case, usability suffers, which has been shown to occur in NICUs among neonatal nurses.¹⁰ However, creating optimal workflows using the heterogeneous collection of today's clinical systems can be time-intensive and

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difficult. At both the infrastructure and user levels, clinical system platforms are cumbersome and often siloed. Clinicians are often met by workflows requiring users to use multiple platforms with different navigation and organizational metaphors, increasing complexity and functionality. In the NICU, this can be experienced by clinicians who may be treating infants with NEC or routinely caring for infants who seem otherwise well for a period of time yet are at risk and can develop NEC suddenly.

Effective team communication when NEC warning signs appear is critical. When treating a new case of NEC, a NICU attending physician or neonatal nurse practitioner (NNP) will need to review the infant's history, physical examination, laboratory, and imaging results. An infant's risk for NEC can be assessed using risk scores like the GutCheck^{NEC} score developed by our team.⁷ Risk factors for NEC arise from diverse vulnerabilities and exposures that occur across the prenatal experience, intrapartum, and NICU clinical course. Quantifying and monitoring clinical adherence to prevention practices is more challenging. Actions that modify or reduce NEC risk are contained in the "NEC-Zero" bundle and prioritize feeding of maternal human milk or donor milk when maternal milk is unavailable; the clinical use of standardized feeding protocols; conservative stewardship of medications like antibiotics and antacids; and use of a score to assess risk and communication changes when symptoms occur (GutCheck^{NEC} with structured Situation-Background-Assessment-Recommendation communication tool). A NEC-Zero adherence score was developed and validated by our team and could support monitoring adherence to prevention practices.¹¹ Supporting team awareness of both risk for NEC and adherence to prevention practices is needed because care is delivered across multiple shifts and by multiple actors who fill different roles yet must coordinate and deliver cohesive patient-centric care.

PURPOSE

The purpose of this study was to describe the clinical workflow and sociotechnical processes that currently exist among clinicians to complete the tasks of NEC risk awareness, timely recognition of symptoms, and treatment for NEC. Our secondary goal was to inform the design of CDS and EHR optimizations for NEC prevention and timely recognition. The institutional review board at the University of Arizona approved and provided oversight for this study.

METHODS

Theoretical Perspective

The theoretical perspective guiding this work integrated Sociotechnical Systems Theory (STS) and the Translating Research into Practice (TRIP) framework from implementation science (see Figure 1, noting that concepts from STS are depicted as circles and TRIP concepts are depicted as boxes).

According to STS, technological characteristics are fit to the social system to optimize an outcome. Key constructs in STS include the client, outcomes, and technology, whereas joint optimization between the social system and the technology is the overarching goal.^{12,13} Joint optimization is necessary for workflow mapping and eventually implementation because it portrays optimal outcomes unattainable without addressing users and the technology. In STS, the social system is considered an open system that adapts. To recognize the influence of technology users interacting in a social system that requires communication processes to move to adoption, the TRIP builds on the diffusion of innovations theory and is well suited for intensive care implementations.^{14,15} Key constructs in the TRIP, developed for implementation research in critical care environments by Titler and Everett, include an innovation (in this case, NEC CDS), communication processes, users, the social system, and adoption (as an outcome). Users are central in implementation science and are depicted centrally in the TRIP model. Titler and Everett reformulated the TRIP from Roger's Diffusion of Innovation theory, which emphasizes that adoption of new innovations differs across users (eg, defined as the early adopter, early majority, late majority, or laggards). In the NEC CDS example, and similar to many system-level CDS interventions, the decision to adopt occurs at the socio-organizational level and individual users do not have full authority to decide if they will adopt the technology, although they choose if, when, and how they use it. Titler and Everett emphasize that the communication processes within the social system are central to engaging users along the continuum of tailoring the innovation to meet users' needs and eventually impact broad adoption (and individual use) across the social system. As an "engaged social process," CDS development and implementation could gather more traction and eventually be better suited to user needs and move toward a system-level outcome (eg, preventing NEC or improving timely recognition of NEC) than if implemented without the social process.

Design

A qualitative descriptive design involving role-based focus groups (FGs) was used to describe workflow processes of NEC recognition for nurses and providers (NNPs, physicians, physician assistants) and to examine how the EHR was being used in the current state. When considering nurses' perspectives, we recognized their role expertise in monitoring for deterioration, implementing the plan of care, notifying providers when concerns of deterioration arose, and taking action when NEC symptoms occur. For providers (ie, attending physicians, NNPs, and physician residents), we examined their workflow in the context of their role. That role may include receiving notifications of concern from another clinician (typically a nurse), completing

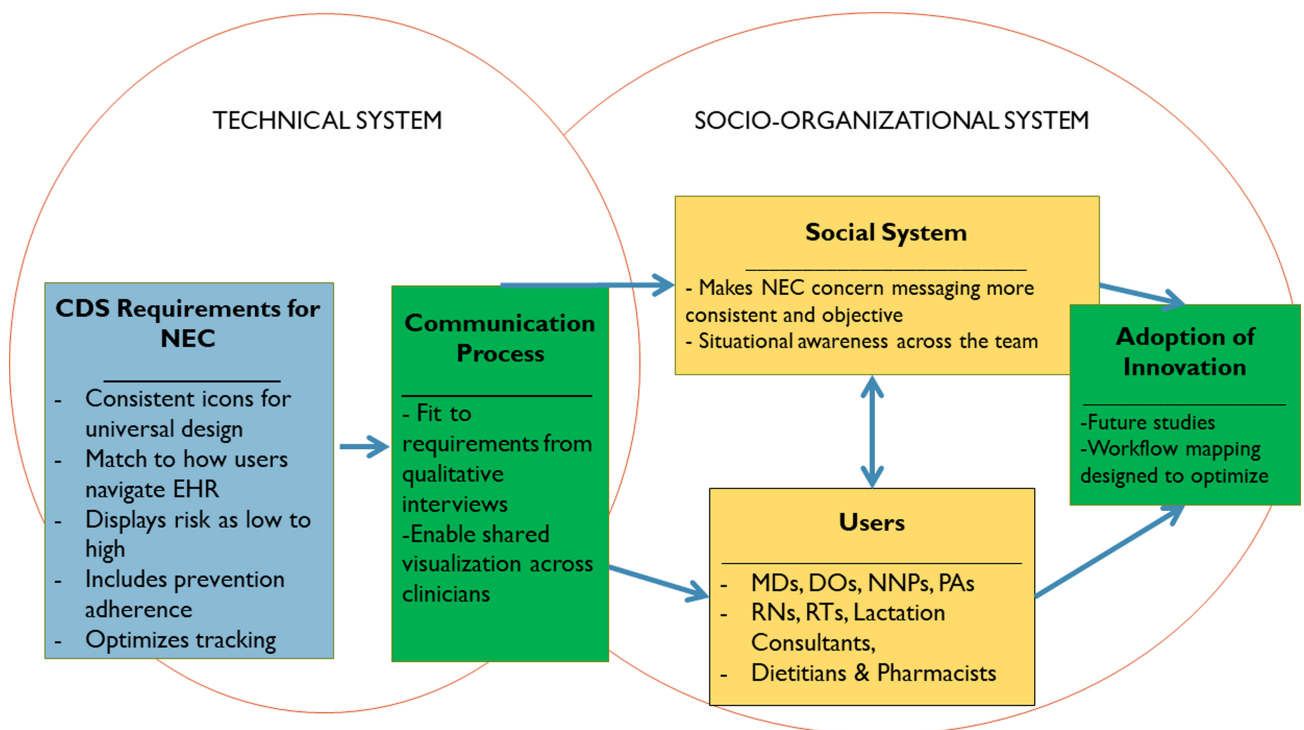


FIGURE 1. Sociotechnical Systems Theory overlaying the TRIP implementation science framework.

their own history and physical examination, ordering diagnostic tests, determining a diagnosis, and making treatment decisions. The FGs provided data to explicate interactions, processes, and workflow within and across roles related to NEC prevention, recognition, and treatments.

Sample and Setting

Participants were recruited from two NICUs in the Southwestern United States. One of the NICUs was a teaching hospital (Unit A), and the second was a community-based high-acuity, high-volume NICU (Unit B). Interviews occurred on-site in the NICUs in a conference room. Clinicians were eligible to participate if they were currently working in the NICU as an RN, physician, NNP, lactation consultant, pharmacist, or dietitian; fluent in English; and worked in the setting for at least 1 year. Different FGs were held for attending neonatologists, nurses (including NNPs), and resident physicians. Site clinical leaders helped the team recruit participants by word of mouth and by sharing recruitment flyers and emails. At Unit A, five FGs were held, followed by six FGs at Unit B.

Study Procedures

After informed consent was obtained, demographic data were collected, and a structured interview guide was used to guide the discussion in the groups. Participants were asked open-ended questions, and follow-up questions were asked to probe their information needs when an infant developed

NEC or had concerning symptoms. Follow-up questions probed their challenges with NEC recognition, asked what tools they used from the EHR to “put the story together,” and explored how they worked as a team. Interviews explored their openness to tools to help them screen NEC risk using CDS. We also asked participants about automation of their current NEC prevention practices and protocols (eg, for feeding, antibiotics, feeding intolerance management, avoidance of antacids, and human milk prioritization). Focus groups were recorded and transcribed verbatim. Immediately after each FG, field notes were taken by the study team. No participant or NICU identifiers were linked to study documents except for the consent. No incentives were given to the participants except for the food offered during the FGs. The study team debriefed after each one to two FGs and after each site's FGs were completed.

Data Analysis

Qualitative content analysis of FG transcripts included mining the transcripts for specific technology features that could support decision-making and then reviewing the transcripts line by line to identify processes of clinician workflow. Workflows were mapped separately for nurses and ordering clinicians (including NNPs, pediatric residents, and attending physicians) by a team of neonatal clinicians. From codes, process terms were mined to identify steps in the process and show clinicians' interaction. Workflow processes were translated into workflow maps

to contextualize and visualize findings. A separate coder with clinical expertise in informatics reviewed transcripts line by line to yield common themes and clinician recommendations on how technology could be redesigned to support their workflow. Categories and a coding scheme were derived inductively from the data transcripts. Coded segments were organized into categories and then into themes.

RESULTS
Sample Characteristics

Eleven FGs were conducted in two NICUs (Unit A, five groups; Unit B, six groups). Unit A was a 30-bed level III unit in an academic medical center. Unit B was an 86-bed level IV unit (per American Academy of Pediatrics NICU classifying criteria) in a non-academic medical center. Both units had facilities to conduct surgery, and pediatric subspecialists were available. Staffing in both units included a mix of clinicians and were affiliated with the same non-profit health system. Clinicians (N = 27) included NICU nurses (37%), physicians (30%), NNPs (19%), and other NICU staff (16%). The average number of years of NICU experience was 16.4 (SD, 13.9), and 81% of respondents were female.

Clinical Workflow Processes for Necrotizing Enterocolitis Recognition

Due to participants' differing clinical roles and responsibilities, two workflow maps were created: one for nurses at the bedside (Figure 2) and another for ordering providers (Figure 3). Nursing workflow for recognition required awareness of risk factors and warning signs; assessment; interpretation; communication; re-interpretation; and taking action. Specifically, this involved processes to consider risk factors (though rarely), identify warning signs, interpret initial signs, and communicate signs to a provider who possessed the authority to order diagnostic testing and initiate treatment. Ordering providers often started their NEC recognition workflow by receiving communication about NEC signs from nursing or another provider (eg, NNP called the resident or the attending physician) and interpreting the level of concern regarding those signs. Characteristics of the social system, such as the professional relationship between individuals, unit culture, time of day, availability, knowledge, and skills, often influenced this step in the workflow process. After receiving communication regarding warning signs of NEC, one of three courses of action is likely, based on the provider's interpretation of level of concern of the clinical findings. Levels of concern are depicted as low (indicating watchful waiting), moderate (accompanied

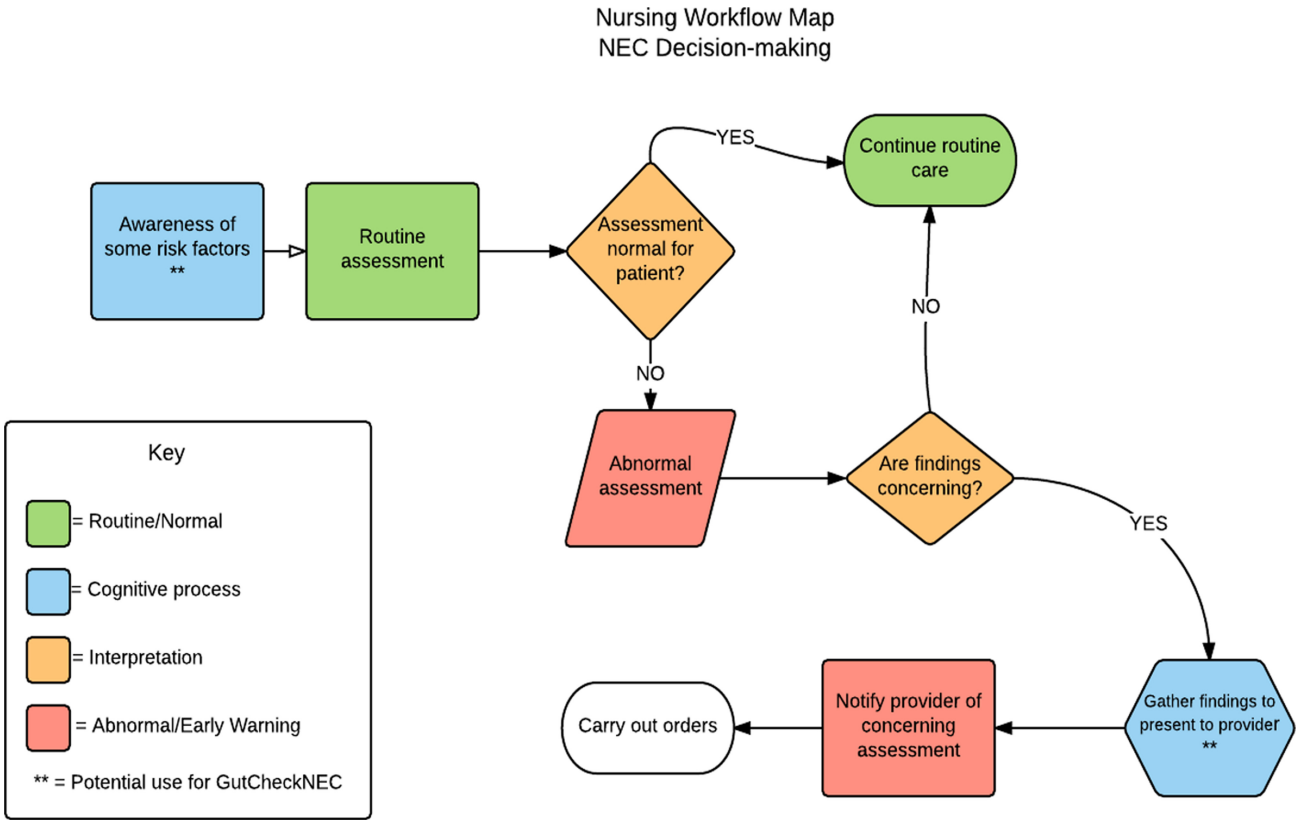


FIGURE 2. Nursing workflows for NEC recognition and action.

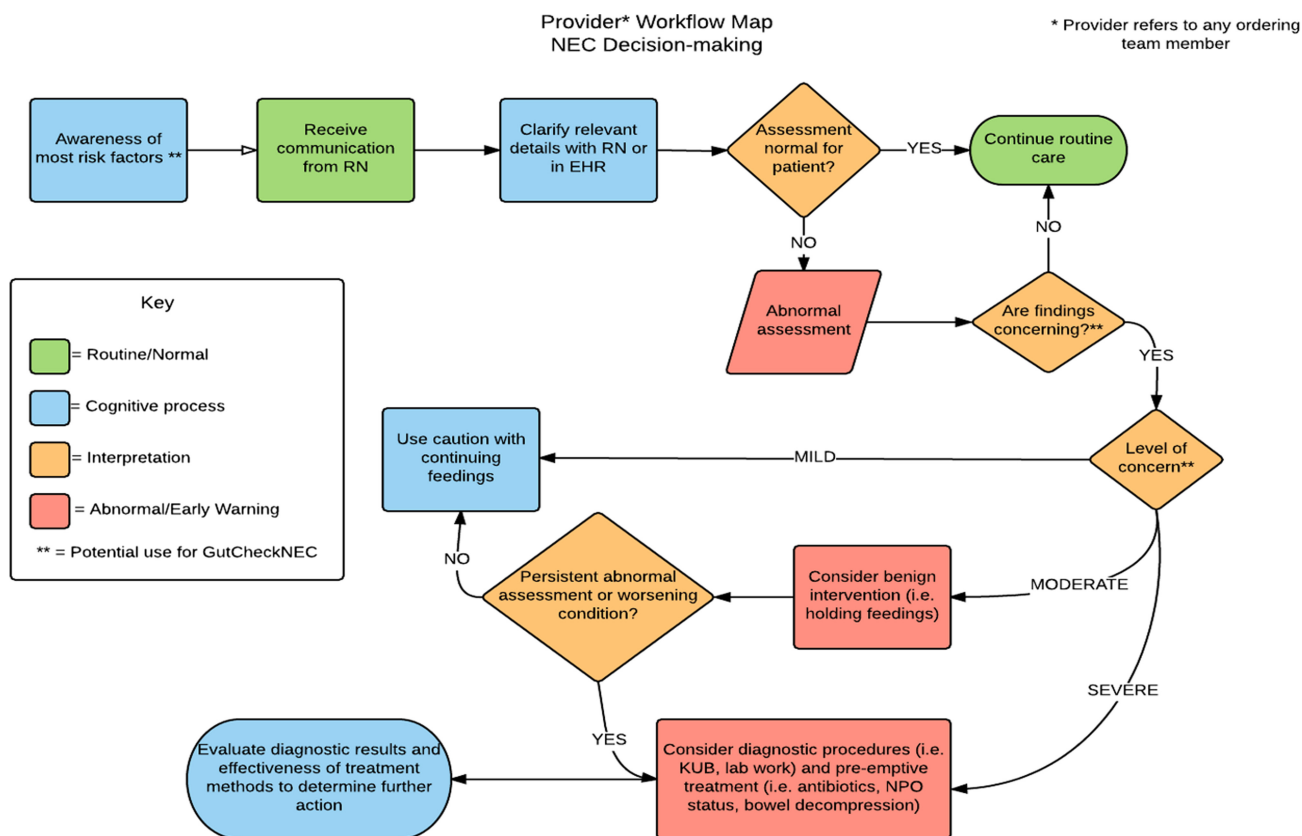


FIGURE 3. Provider workflows for NEC recognition and action.

by withholding feedings and obtaining an abdominal x-ray), and serious (indicating urgent diagnostic testing and rapid initiation of a therapeutic regimen).

Themes that arose around NEC recognition included inputs and actions to (1) identify signs, (2) interpret signs, (3) communicate signs by notifying provider, (4) social system influences that altered provider interpretation and response to notification, and (5) practice variations. Coded statements suggested that social systems and structures within NICUs such as clinician hierarchies and confidence in the team influence timeliness of care. Due to mistrust and ambiguous messaging, slower response led to altered (often slower) NEC recognition and potentially worsening NEC. Notably, in a teaching hospital, the person who often noticed warning signs (ie, the RN) communicated through two people (resident and/or NNP) before notifying the attending physician. The attending physicians rarely to never received notifications directly from the bedside nurse in the teaching hospital that included resident physicians and NNPs.

Design Expectations for Necrotizing Enterocolitis-Related Clinical Decision Support

Two themes emerged reflecting the needs and wants of clinicians caring for infants at risk for or who have developing

NEC: (1) preferred features of NEC CDS and (2) EHR system optimization. Themes addressed preferences for (1) a clinical dashboard, (2) new CDS to integrate risk scoring and early warning, (3) breast milk administration tracking, (4) NEC reporting and surveillance monitoring, and (5) local EHR optimization to reduce data entry and facilitate standardized ordering (Table 1).

Clinical Dashboard

Clinicians desired a single, central location to access aggregated data relevant to NEC prevention, risk awareness, and recognition. By culling clinician narratives about data used to assign risk, identify signs, and interpret signs, several specific data items to populate a dashboard were identified. Clinicians discussed their need for assessment, background (eg, maternal and clinical course history), feeding history, laboratory, and diagnostic imaging data. Participants, especially nurses, said that structuring the display of data into a NEC dashboard would support clear communication with providers. Both groups identified that they needed to see the same information that the other team member was viewing (which is not the case because restrictions are placed on data viewing and access to different parts of the record). Nurses described that “making a case” to the provider required them to describe

Table 1. Qualitative Findings Informing NEC-Zero Clinical Decision Support Needs and Wants

Category	Significant Supporting Statement
Theme 1: Technical requirements for NEC clinical decision support	
Clinical dashboard	- "There's definitely opportunity for change [in EHR]." - "[To gather information] I would have to look in about 3 different places...click a lot but buttons to get that for you... it seems like it would be very simple to create just a section where all of that information [NEC risk scoring, clinical signs, feeding information, exposure to preventive treatments] flowed over into so it would give a much quicker snapshot." - "I think it would be helpful if the record would truly pull some information together that was accurate...like historical information like mom's history...and maybe what is this baby at risk for."
Clinical decision support	- "If there was a program that could go in and pull that information out (predetermined data) so maybe we have a program but doesn't know to recognize these and then an alert * like the kid has 4 of our * and it * automatically. That would be something that even if they did have risk factors that have popped up every day."
Breast milk administration tracking	- "We don't document mom's pumping volumes or mom's frequency of pumping. It would be so nice—it seems like we're good about that." - "We will kind of see oh, we're running out of milk. We're going to have to use formula. That kind of triggers to ask mom why, are you still pumping? Because sometimes they just won't be bringing their milk in."
Reporting and surveillance tool	- "Risk identification that is easy to view for all staff, not just bedside nurses. Identifying babies' risk and warning signs regularly monitored by everyone responsible for baby." - "Sharing information about rates, prevalence, what we've done recently showing recently whether it's graphs, just statistics and being able to point to something and say starting at this point when this was rolled out here's the change."
Theme 2: EHR system optimization	
Communication	- "If you can easily find everything in one place—I think it would make it a lot faster, or when you're charting." - "We've talked about actually having an SBAR in general. We're trying to accumulate that through our RN-MD communication. They're actually coming out with an SBAR tool for report so that it's more thorough and also using SBAR when you call the doctors."
Family education	- "With the million new pieces of information they're getting, I don't know how effective any sort of teaching...is going to be."
Interoperability of records	- "Based on the record and what transferred, whatever they had done there, we wouldn't repeat obviously if they did eye care and vitamin K and all that."
Documentation	- "We ask so much of these nurses. Not only to take care for these babies but to chart logs—amounts of information in these EHRs. Redundant charting." - "Yeah, I mean especially if it's something that can automatically populate over and just be left somewhere obvious for the nurses to look at."
Order sets	- "They can click the orders that would already be pre-populated."

symptoms within a context that could allow the provider to distinguish between different disease processes. However, some mentioned that they did not always know what was evolving but knew something “wasn't right” or that symptoms were not specific. One participant described making the case with “...belly distended, belly hard, belly discolored, kid is having brady's and desats, high residuals, girth's going up, color looks like crap—just that septic look to them, they're just not acting right, they're more lethargic.... You really have to present your case.”

Clinical Decision Support

A CDS that intelligently filters and presents NEC-specific information was another emergent theme. A CDS may include computerized alerts, reminders, order sets, or clinicians' guidelines that enhance clinical decision-making, for instance, incorporating GutCheck^{NEC} into an alert, creating NEC-specific order sets, and firing an alert to the RN and provider

based on risk status and warning signs. Currently, physicians reported low use of the nursing documentation because of difficulty retrieving data.

Breast Milk Administration Tracking

Participants noted a gap in the ability to record milk administered, volume tracking, and free text comments associated with feeding. A solution is a tool to ensure the tracking and documenting of the amount of milk, provides support for labeling, and tracks storage. This solution increases patient safety and supports nursing workflow from receiving the order to administering and documenting the feeding.

Optimizations for Reporting and Communication

Reporting and surveillance tools such as the ability to pull reports on NEC rates and prevalence and informational summaries to monitor NEC performance to support NEC was another emerging theme from FG interviews. Several

sub-themes emerged for potential optimization projects: (1) communication and documentation, (2) order sets, (3) family education, and (4) interoperability of records. Participants desired better communication with their current systems, including provider handoff notification and documentation and enhanced documentation. Enhanced documentation included creating documentation templates that aggregate useful data during provider handoff without navigating the chart endlessly for information. Participants noted sub-optimal experiences with electronic documentation, particularly redundant or duplicative charting. Recommendations included the utilization of the last charted values and auto-population of documentation values that share the elements.

Creating order sets such as feeding and NEC-diagnosis-specific orders was determined an optimization. Order set recommendations included evidence-based orders related to diagnostic tests, assessments, medications, consults, and so forth. Participants recommended adding NEC order sets to save time and improve consistency when advancing feedings or ordering transfusions,

Participants noted that effective and innovative ways to educate parents and family members regarding NEC are important optimization projects. Participants were concerned that parents are commonly overloaded with information, education was sometimes ineffective, and supporting parents' information needs was time-consuming. Education optimization using interactive patient engagement is needed. Additionally, when an infant is transferred between hospitals, the value of information sharing between the sites is paramount and especially fragmented when the sites are from different organizations. When surgery for NEC is needed, this challenge is urgent and fragmented information sharing leaves parents as the primary information source for the receiving hospital. With optimized interoperability, a complete and current record supports clinical decision-making across the disease trajectory while diminishing the information burden for parents.

DISCUSSION

To our knowledge, this is the first project to examine what is needed to support clinical decision-making for NEC risk awareness, timely recognition, and action when NEC concerns arise. Few other studies have examined CDS design or use in the NICU environment.¹⁶ Our findings highlight the need for EHR optimization and echo other research that demonstrates a difficult documentation and information retrieval experience for NICU RNs.¹⁰ Gaps in electronic-mediated communication among clinicians were shown, as physicians found the nursing documentation to be “onerous” and “hard to retrieve”—a challenge that has been shown in other patient care units besides NICU.^{17,18}

Strengths of this work involve our human-centered approach that integrates perspectives of those who assess, mon-

itor, and identify NEC concerns (ie, bedside RNs) and those who receive messages of concern and act through ordering and diagnostics (ie, providers including MDs, Doctor of Osteopathic Medicine, NNP). This study's results are consistent with Ozkaynak et al's¹⁹ recommendations that CDS should be dependent on a specific clinical workflow, allow for multiple roles, and be situation-specific to affect behavior change. Similar to Ozkaynak et al's findings that different clinicians take different steps and make decisions at different points within care delivery, our results showed two distinct workflows for RNs at the bedside and for providers who are notified of NEC concern by the RNs or identify signs of deterioration through their own assessment (rare). By accounting for clinicians' workflow and processes, teams can inform, develop, refine, and sustain effective CDS, whether focused on pediatric emergency departments and antibiotic stewardship¹⁹ or on NEC prevention and timely recognition.

To accomplish our goal to design CDS that would be operable across sites, we completed FGs at a teaching institution and a non-teaching institution. This approach is similar to that taken by Orenstein and colleagues,²⁰ who examined workflows across organizations to enable a flexible CDS solution that could work across systems. To support broader use, we have developed a repository of logic diagrams and workflows to support local adaptations of the CDS for NEC to design a standalone CDS that can integrate locally but resides external to the EHR (contact authors for more information).

Limitations

Limitations of our approach included recruiting clinicians from two units and the time intensity of data analysis. Unobtrusive methods using audit logs to assess optimal navigation patterns or other workflow analysis techniques could augment our approach in future studies.²¹ We recognize that many workflow studies involve observations that can be augmented by interviews or log file review. It is important to know that no CDS for nursing relative to NEC was in use at these two units and is extremely rare nationally. To our knowledge, none exists. Within our study constraints, we were unable to observe actual clinical workflow. With that in mind, the workflows we modeled from FGs need validation in the future with robust clinical observation. Similarly, the CDS components we propose may vary subject to unique social systems and technical constraints operating in each NICU environment. Our effort to collect data at two different types of NICUs was intentional to examine the various types of communication patterns at the different sites. Similarly, it may be that the processes we outlined differ based on the socio-organizational culture and technical constraints (eg, level of EHR optimizations, tolerance for CDS customization, presence of clinical champions, and skilled technology professionals to work together on CDS). With these many factors in mind,

there may be a place to design and implement NEC CDS as a module that does not have to be re-built in every new system (eg, as a SMART on FHIR tool).

Capturing the workflow at a granular and more objective level could strengthen our conclusions but was not feasible in this study. As such, we represented the workflow by group, which minimizes our ability to handle individual variations. However, technology is designed for the group and not the individual in EHR instances, making our approach translatable to actions that can be taken to design new technologies. Study participants were familiar with Cerner and/or Epic EHRs. The degree of EHR customization and optimization varies across system and cross-organization, which dramatically impacts the clinician experience.^{10,22} Finally, parents were not included as participants, which could limit the usefulness of these findings for them or for organizations that wish to include parents as primary decision-makers.

Implications for Practice and Research

Electronic health record optimization and communication improvements are needed to foster timely and accurate NEC recognition and prevention, perhaps in the form of CDS. Family support materials to foster engagement are available (see NEC Society.org and Neczero.nursing.arizona.edu). A dashboard approach is needed to support clinical decision-making. Once CDS is available and implemented for NICUs in the field of NEC care, it will need to be rigorously evaluated. This can be accomplished locally, within individual health systems and NICUs, or across units. Strategies to overcome barriers to CDS implementation can be informed by implementation science, particularly the TRIP framework that emphasizes the central role of users, communication processes, and the social system within which adoption occurs.

CONCLUSION

Improving early recognition of NEC requires closing the gap between recommended care and existing practice. Workflow maps illuminate nursing interpretation and decision-making tasks amenable to CDS, especially in the context of assessing for NEC risk in the context of emerging symptoms to support clear communication with other members of the care team. Maps demonstrate the need to involve the team of clinicians and the critical nature of effective communication throughout the process of NEC prevention and recognition.

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